

Importance Sampling

Monte Carlo Statistical Methods

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1 Goal and Setup

We want to estimate the probability of a rare event under a standard normal random variable

$$X \sim \mathcal{N}(0, 1).$$

Consider the tail event

$$A = \{x \geq 4.5\}.$$

The probability of interest is

$$p = \mathbb{P}(X \geq 4.5) = \int_{4.5}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx.$$

Using symbolic integration (for example with SymPy), we obtain the numerical value

$$p = 3.39767312476808 \times 10^{-6}.$$

This is a very small number: the event happens roughly three to four times in a million independent standard normal draws.

Our goals in this note are:

- to define the plain Monte Carlo estimator and the importance sampling estimator for this probability,
- to derive the expectation and variance of the plain Monte Carlo estimator,
- to use the central limit theorem (CLT) to understand how many samples plain Monte Carlo needs,
- to see why that sample size is astronomically large for this rare event,
- to introduce an importance sampling estimator with a better proposal distribution,
- to derive and interpret the variance of the importance sampling estimator and explain why it can be much smaller than the Monte Carlo variance.

Throughout, we will use the notation

- $\mathbb{E}_f$ for expectations when $X \sim f$ (the target standard normal density),
- $\mathbb{E}_g$ for expectations when $X \sim g$ (the proposal density),
- $\hat{p}_n^{MC}$ for the Monte Carlo estimator,
- $\hat{p}_n^{IS}$ for the importance sampling estimator.

2 Analytical Value of the Rare Event Probability

Let

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

denote the standard normal density.

Then the tail probability is

$$p = \int_{4.5}^{\infty} \phi(x) dx = 3.39767312476808 \times 10^{-6}.$$

This value is treated as the ground truth. Every estimator we build is trying to recover this number from random samples.

3 Plain Monte Carlo Estimator

3.1 Definition

Let

$$X \sim f = \mathcal{N}(0, 1).$$

We can write the probability as an expectation under f :

$$p = \mathbb{P}_f(X \geq 4.5) = \mathbb{E}_f[\mathbb{1}_{\{X \geq 4.5\}}].$$

Define the Bernoulli random variable

$$Y = \mathbb{1}_{\{X \geq 4.5\}} \in \{0, 1\}.$$

Then

$$\mathbb{E}_f[Y] = p, \quad \text{Var}_f(Y) = p(1 - p).$$

Given i.i.d. draws

$$X_1, \dots, X_n \sim f,$$

we define the Monte Carlo estimator

$$\hat{p}_n^{MC} = \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{\{X_i \geq 4.5\}} = \frac{1}{n} \sum_{i=1}^n Y_i.$$

3.2 Expectation and Variance of $\hat{p}_n^{MC}$

Unbiasedness:

$$\begin{aligned} \mathbb{E}_f[\hat{p}_n^{MC}] &= \mathbb{E}_f \left[\frac{1}{n} \sum_{i=1}^n Y_i \right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}_f[Y_i] \\ &= \frac{1}{n} \cdot n \mathbb{E}_f[Y] = p. \end{aligned}$$

So $\hat{p}_n^{MC}$ is an unbiased estimator of p :

$$\mathbb{E}_f[\hat{p}_n^{MC}] - p = 0.$$

Variance:

$$\begin{aligned}\mathrm{Var}_f(\hat{p}_n^{MC}) &= \mathrm{Var}_f\left(\frac{1}{n} \sum_{i=1}^n Y_i\right) \\ &= \frac{1}{n^2} \sum_{i=1}^n \mathrm{Var}_f(Y_i) = \frac{1}{n^2} \cdot n p(1-p) \\ &= \frac{p(1-p)}{n}.\end{aligned}$$

The Monte Carlo standard error is therefore

$$\sqrt{\mathrm{Var}_f(\hat{p}_n^{MC})} = \sqrt{\frac{p(1-p)}{n}}.$$

For a rare event with very small p , this variance will be roughly p/n , which is tiny numerically but problematic because almost all of the individual samples are zeros.

3.3 Numerical Result for $n = 100,000$

Consider the experiment with $n = 100,000$ samples from the standard normal:

```
num_samples = 100000
standard_normal_sampler = np.random.randn

def g_evaluator(x: np.ndarray) -> np.ndarray:
    return x > 4.5

Pz_mc_estimate = monte_carlo(num_samples, standard_normal_sampler,
                              g_evaluator, cumsum=False)
print(f"MC estimate with {num_samples:,} samples = {Pz_mc_estimate.estimate:5E}")
print(f"Truth = {Pz:5E}")

# Output:
# MC estimate with 100,000 samples = 0.000000E+00
# Truth = 3.39767312476808E-6
```

Numerically we obtain

$$\hat{p}_{100000}^{MC} = 0, \quad p = 3.39767312476808 \times 10^{-6}.$$

This is not a numerical bug. For this event, p is so small that with 100,000 standard normal samples we expect less than one hit on average:

$$np = 100,000 \times 3.4 \times 10^{-6} \approx 0.34.$$

So it is more likely than not that we see no hits at all and the estimator outputs zero.

4 How Many Samples Does MC Need? CLT and Relative Error

4.1 CLT for the Bernoulli Average

Recall:

$$Y_i = \mathbb{1}_{\{X_i \geq 4.5\}}, \quad X_i \sim f, \quad Y_i \sim \text{Bernoulli}(p),$$

and

$$\hat{p}_n^{MC} = \frac{1}{n} \sum_{i=1}^n Y_i.$$

We have

$$\mathbb{E}_f[Y_i] = p, \quad \text{Var}_f(Y_i) = p(1-p).$$

By the central limit theorem (CLT),

$$\sqrt{n} \frac{\hat{p}_n^{MC} - p}{\sqrt{p(1-p)}} \xrightarrow{d} N(0, 1),$$

so for large n , the standardized error

$$\frac{\hat{p}_n^{MC} - p}{\sqrt{p(1-p)/n}}$$

is approximately standard normal.

4.2 Relative Error and Confidence Level

We want to control the relative error:

$$\frac{|\hat{p}_n^{MC} - p|}{p} \leq \varepsilon \quad \Longleftrightarrow \quad |\hat{p}_n^{MC} - p| \leq \varepsilon p,$$

for some tolerance $\varepsilon > 0$.

We also want this to hold with confidence $1 - \alpha$ (for example $1 - \alpha = 0.95$). Let $q_{1-\alpha/2}$ denote the $(1 - \alpha/2)$ quantile of the standard normal distribution. For 95% confidence,

$$\alpha = 0.05, \quad q_{0.975} \approx 1.96.$$

Under the CLT approximation,

$$\mathbb{P} \left(\left| \frac{\hat{p}_n^{MC} - p}{\sqrt{p(1-p)/n}} \right| \leq q_{1-\alpha/2} \right) \approx 1 - \alpha.$$

To enforce the relative error constraint within this probability, we require

$$\frac{\varepsilon p}{\sqrt{p(1-p)/n}} = q_{1-\alpha/2}.$$

This equation gives us a design rule for n .

4.3 Solving for n

Starting from

$$\frac{\varepsilon p}{\sqrt{p(1-p)/n}} = q_{1-\alpha/2},$$

square both sides:

$$\frac{\varepsilon^2 p^2}{p(1-p)/n} = q_{1-\alpha/2}^2.$$

Rearrange to isolate n :

$$\varepsilon^2 p^2 \cdot \frac{n}{p(1-p)} = q_{1-\alpha/2}^2 \implies n = \frac{q_{1-\alpha/2}^2 p(1-p)}{\varepsilon^2 p^2} = \frac{q_{1-\alpha/2}^2 (1-p)}{\varepsilon^2 p}.$$

For a rare event with $p \ll 1$, we can approximate $(1-p) \approx 1$, so

$$n \approx \frac{q_{1-\alpha/2}^2}{\varepsilon^2 p}.$$

This shows two things:

- The sample size grows like $1/(\varepsilon^2 p)$ as p becomes small.
- For very small p , the required number of samples explodes.

4.4 Numerical Example: 10% Relative Error at 95% Confidence

Now plug in the specific values

$$p = 3.39767312476808 \times 10^{-6}, \quad \varepsilon = 0.1, \quad 1 - \alpha = 0.95, \quad q_{0.975} = 1.96.$$

Then

$$n \approx \frac{1.96^2}{(0.1)^2 p} = \frac{3.8416}{0.01 p} = \frac{384.16}{p}.$$

Substituting our p :

$$n \approx \frac{384.16}{3.39767312476808 \times 10^{-6}} \approx 1.13 \times 10^8.$$

So we need roughly

$$n \approx 113 \text{ million samples}$$

to get 10% relative error with 95% confidence using plain Monte Carlo.

4.5 Interpretation: How Often Do We See a One?

The true probability is

$$p = 3.39767312476808 \times 10^{-6}.$$

On average, we see one success (one hit in the tail) every

$$\frac{1}{p} \approx 294,300$$

standard normal samples. That means:

- In a block of about 300,000 samples, we typically see one 1 and almost 299,999 zeros.
- To build a stable estimator with small relative error, we need many such successes, not just one or two.
- The CLT calculation tells us that to have enough successes to shrink the relative error to 10%, we should plan for on the order of 10^8 samples.

In practice, this is not just a large number. For a realistic simulation where each sample may involve a physics engine, a robot dynamics model, or a complicated likelihood evaluation, 113 million evaluations is simply too expensive.

The Monte Carlo estimator remains unbiased, but its convergence is squeezed by the fact that almost every sample contributes a zero to the estimator. The zeros dominate, the rare ones are too infrequent, and the relative error decays painfully slowly.

This is the core failure mode of plain Monte Carlo for rare events: the distribution we sample from almost never visits the event we care about.

5 Importance Sampling Estimator

To fix this, we change *where* we sample from. Instead of drawing X from f and almost never seeing the event A , we draw X from a proposal distribution g that places most of its mass inside A .

5.1 Choice of Proposal

We choose g to be an exponential distribution shifted to start at 4.5:

$$g(x) = \begin{cases} e^{-(x-4.5)}, & x \geq 4.5, \\ 0, & x < 4.5. \end{cases}$$

Under this proposal, almost every sampled X lies in the region $x \geq 4.5$, so the rare event is not rare under g .

5.2 Rewriting the Probability Under g

We can rewrite p as an expectation under g using the identity

$$f(x) dx = \frac{f(x)}{g(x)} g(x) dx$$

whenever $g(x) > 0$. For our indicator integrand,

$$p = \int_{4.5}^{\infty} \phi(x) dx = \int_{4.5}^{\infty} \mathbb{1}_{\{x \geq 4.5\}} \phi(x) dx = \int_{4.5}^{\infty} \mathbb{1}_{\{x \geq 4.5\}} \frac{\phi(x)}{g(x)} g(x) dx.$$

Define the weight

$$w(x) = \frac{f(x)}{g(x)} = \frac{\phi(x)}{g(x)},$$

for $x \geq 4.5$. Then

$$p = \mathbb{E}_g[\mathbb{1}_{\{X \geq 4.5\}} w(X)],$$

where $\mathbb{E}_g$ denotes expectation when $X \sim g$.

5.3 Definition of $\hat{p}_n^{IS}$ and Unbiasedness

Given i.i.d. draws

$$X_1, \dots, X_n \sim g,$$

the importance sampling estimator is

$$\hat{p}_n^{IS} = \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{\{X_i \geq 4.5\}} w(X_i).$$

Unbiasedness:

$$\begin{aligned} \mathbb{E}_g[\hat{p}_n^{IS}] &= \mathbb{E}_g \left[\frac{1}{n} \sum_{i=1}^n \mathbb{1}_{\{X_i \geq 4.5\}} w(X_i) \right] \\ &= \frac{1}{n} \sum_{i=1}^n \mathbb{E}_g [\mathbb{1}_{\{X_i \geq 4.5\}} w(X_i)] \\ &= \mathbb{E}_g [\mathbb{1}_{\{X \geq 4.5\}} w(X)] \\ &= p. \end{aligned}$$

So $\hat{p}_n^{IS}$ is also an unbiased estimator of p .

5.4 Variance of $\hat{p}_n^{IS}$

Let

$$Z = \mathbb{1}_{\{X \geq 4.5\}} w(X), \quad X \sim g.$$

Then

$$\hat{p}_n^{IS} = \frac{1}{n} \sum_{i=1}^n Z_i, \quad Z_i \text{ i.i.d. copies of } Z.$$

We already know

$$\mathbb{E}_g[Z] = p.$$

The variance of a single Z is

$$\text{Var}_g(Z) = \mathbb{E}_g[Z^2] - (\mathbb{E}_g[Z])^2 = \mathbb{E}_g[\mathbb{1}_{\{X \geq 4.5\}} w(X)^2] - p^2.$$

Since $\hat{p}_n^{IS}$ is an average of n i.i.d. copies of Z ,

$$\text{Var}_g(\hat{p}_n^{IS}) = \frac{1}{n} \text{Var}_g(Z) = \frac{1}{n} (\mathbb{E}_g[\mathbb{1}_{\{X \geq 4.5\}} w(X)^2] - p^2).$$

This expression shows clearly what matters for importance sampling variance: the second moment of the weighted indicator under g . If $w(X)$ is well-behaved (not too large, not too variable) on the region where the indicator is 1, then the variance can be much smaller than the Monte Carlo variance.

6 MC vs IS: Variance Comparison and Intuition

We now compare the variances of the two unbiased estimators.

For plain Monte Carlo:

$$\text{Var}_f(\hat{p}_n^{MC}) = \frac{p(1-p)}{n}.$$

For importance sampling:

$$\text{Var}_g(\hat{p}_n^{IS}) = \frac{1}{n} (\mathbb{E}_g[\mathbb{1}_{\{X \geq 4.5\}} w(X)^2] - p^2).$$

Both share the factor $1/n$, as they are averages of n i.i.d. terms. The difference is completely inside the expectations:

- Monte Carlo variance depends on $p(1-p)$.
- Importance sampling variance depends on $\mathbb{E}_g[\mathbb{1}_{\{X \geq 4.5\}} w(X)^2]$.

The estimator with the smaller variance for a given n is the more efficient one, since both are unbiased.

6.1 Why MC is Bad Here

For plain MC:

$$\text{Var}_f(\hat{p}_n^{MC}) = \frac{p(1-p)}{n} \approx \frac{p}{n},$$

because p is tiny and $1-p \approx 1$.

With $p \approx 3.4 \times 10^{-6}$, this gives

$$\text{Var}_f(\hat{p}_n^{MC}) \approx \frac{3.4 \times 10^{-6}}{n}.$$

The key issue is not just the numerical value of the variance, but the structure of the estimator: almost all Y_i are zero, and the rare ones are so infrequent that many independent blocks of size n will contain no hits at all. The estimator is unbiased in the long run, but in finite samples it is dominated by zeros and has huge relative fluctuations.

6.2 Why IS Can Be Much Better

Under our exponential proposal g , the random variable X lives almost entirely in the tail region $x \geq 4.5$. The indicator $\mathbb{1}_{\{X \geq 4.5\}}$ is then identically 1 (up to numerical underflow), so the randomness in Z comes almost entirely from the weights $w(X)$.

In other words:

- Plain MC: randomness comes from whether we even see the rare event or not. Most samples are zero.
- IS: randomness comes from fluctuations in $w(X)$ inside the tail region. Every sample contributes directly to the estimate.

If the proposal g is chosen so that its shape in the tail is roughly aligned with f (for example, matching decay rates), then $w(X)$ stays moderate and

$$\mathbb{E}_g[w(X)^2 \mathbb{1}_{\{X \geq 4.5\}}]$$

does not explode. In that case, for a fixed n ,

$$\text{Var}_g(\hat{p}_n^{IS}) \ll \text{Var}_f(\hat{p}_n^{MC}).$$

Intuitively:

- Under f , the rare event is almost never seen. We are averaging mostly zeros and occasionally a one.
- Under g , the rare event is seen in almost every sample, and we adjust by weights so that the expectation remains correct.

Instead of waiting $1/p \approx 294,300$ samples to see a single useful hit, importance sampling turns almost every sample into useful information about the quantity we care about. The weights play the role of correcting the bias introduced by this aggressive oversampling of the tail.

7 Numerical Result with Importance Sampling

We now implement the importance sampling estimator using the shifted exponential proposal and compare it to the analytical truth.

```
num_samples = 100000

def proposal_sampler(n: int) -> np.ndarray:
    return stats.expon.rvs(size=n, loc=4.5, scale=1)

def proposal_pdf(x: np.ndarray) -> np.ndarray:
    return stats.expon.pdf(x, loc=4.5, scale=1)

def target_pdf(x: np.ndarray) -> np.ndarray:
    return stats.norm.pdf(x)

Pz_is = importance_sampling(num_samples,
                           proposal_sampler,
                           proposal_pdf,
                           target_pdf,
                           g_evaluator,
                           False)

print(f"Important Sampling estimate with {num_samples:,} samples = "
      f"{Pz_is.estimate_is:5E}")
print(f"Truth = {Pz:5E}")

# Output:
# Important Sampling estimate with 100,000 samples = 3.390223E-06
# Truth = 3.39767312476808E-6
```

With $n = 100,000$ samples from the proposal g , we obtain:

$$\hat{p}_n^{IS} \approx 3.390223 \times 10^{-6}, \quad p = 3.39767312476808 \times 10^{-6}.$$

Compare this with plain Monte Carlo using the same 100,000 samples from f :

$$\hat{p}_n^{MC} = 0.$$

For the same computational budget ($n = 10^5$ sample evaluations),

- plain Monte Carlo completely misses the rare event and returns zero,
- importance sampling produces an estimate extremely close to the analytical truth.

The difference is not a small constant factor. Plain Monte Carlo needs on the order of 113 million samples to achieve 10% relative error with 95% confidence. Importance sampling, with a well-chosen proposal, achieves a good approximation to the true rare-event probability with only 100,000 samples.

This is exactly why importance sampling is essential in rare event estimation: it changes the sampling distribution so that the event of interest is frequently observed, and then uses weights to correct back to the original target distribution.